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RESEARCH ARTICLE

MEDICAL PHYSICS

Heavy metal in radiation therapy: A simple method to differentiate hip prosthesis materials

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Abstract

Background: In recent years, the number of hip replacement patients receiving radiation therapy has steadily increased. In parallel, strategies have been developed to reduce metal artifacts in computed tomography (CT) images and improve the accuracy of dose calculation algorithms. However, in certain situations, knowledge of the type of prosthesis material is required to accurately determine the dose distribution.

Purpose: This study aims to identify physical materials in hip prostheses to correctly assign them in the treatment planning system and improve dose calculation accuracy.

Methods: We first verified the validity of the extended CT mass density calibration curve measured on titanium (Ti) and stainless steel (SS) metal inserts of two different diameters. Then using dedicated reference objects of various circular diameters, we developed a method based on interpolation functions to differentiate between Ti and SS material groups. Forty data sets from 18 patients were used to validate our method on two different reconstruction kernels: a standard Br44f and the electron DirectDensity (Sd40f) kernels from Siemens.

Results: Hounsfield units (HU) of Ti and SS inserts were found to vary widely depending on insert diameter, CT spectrum, and reconstruction kernels due to cupping artifacts. The largest HU difference (-79%) was obtained for SS at 70 kV with Br44f when the diameter increased from 8 to 30 mm. Therefore, under these conditions, the extended CT-density calibration curve is not recommended for heavy metal density determination. Using our interpolation-based method, we achieved excellent detection (100%) and material differentiation (100%) results for stems in both reconstruction kernels. At CT energies between 110 and 140 kV, the detection and material differentiation rates were 93.3% and 92.9% for the heads and 93.3% and 92.9% for the acetabular cups, respectively, with the Br44f. Similarly, the use of Sd40f resulted in detection and differentiation rates of 94.7% and 100% for the heads and 100% and 95.0% for the acetabular cups, respectively.

Conclusion: This method makes it possible to differentiate between hip prosthesis materials and correctly assign them to the Ti or SS group without prior knowledge of the prosthesis type, regardless of the reconstruction kernels. In combination with the Acuros XB (Varian) or Monte Carlo dose algorithms, excellent dosimetric accuracy can be achieved even in the vicinity of hip prostheses. By performing basic measurements, the method can be adapted to other CT units and reconstruction kernels, replacing the use of an extended CT-density calibration curve.

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KEYWORDS
Acuros XB, calculation dose, calibration curve, ceramic, DirectDensity, high-density material, hip prosthesis, metal, stainless steel, titanium

1 | INTRODUCTION

Hip replacement, a surgical procedure that replaces the hip joint with a prosthetic implant, is one of the most commonly performed surgeries in the world. In 2021, Switzerland had the highest rates of hip replacement among the Organization for Economic Co-operation and Development (OECD) countries with available data.[1] In the US, an increase in primary total hip arthroplasty (THA) of +129% and +284% is expected for 2030 and 2040, respectively.[2] The main indication for joint replacement surgery is osteoarthritis, which has increased by +113% worldwide since 1990. Although hip replacement surgery is most commonly performed on people over the age of 60, it can also be performed on younger people, usually to treat severe physical damage to the joint caused by a hip fracture.[1]

Meanwhile, the incidence of cancer continues to increase. In 2020, there were 19.3 million new cases and 10 million cancer deaths worldwide.[3] However, if the incidence remains stable and population growth and aging continue with recent trends, an increase of 47% in new cases for 2040 is expected. In Europe, this value is expected to increase by nearly 21%.[4] The need for cancer treatment to include radiotherapy is now widely accepted, and more than 50% of cancer patients require radiotherapy as part of their cancer care.[5,6] In this context, the number of patients with prostheses who will benefit from radiotherapy is likely to continue to increase over the next few years.

In 2003, the AAPM TG-63 published a dosimetric recommendation for patients with hip prostheses undergoing pelvic irradiation[7] and Wieslander et al. showed that for accurate dose determination by TPS, the composition and design of the implant must be known.[8] Various strategies have been developed to avoid irradiating prostheses.[9–11] Although rotational treatments such as volumetric-modulated arc therapy (VMAT) have become commonplace[12] and are usually based on avoidance sectors,[13] knowledge of the prosthesis material properties is necessary during the dosimetry stage for modern algorithms. Indeed, treatment planning systems (TPS), which use Acuros XB (AXB) (Varian Medical Systems Inc., Palo Alto, California, USA) and Monte Carlo (MC)-based algorithms, require the type of material to produce a dose calculation.[14]

For typical materials (from air to hard bone), this information is provided by the physical or electronic CT-density curve[15] or the stoichiometric calibration curve.[16] However, for very high-density materials such as metals, attenuation, artifacts, and data truncation produce uncertainties in addition to the fact that the HU values of these materials are highly dependent on the CT energy spectrum.[17]

Among recent CT scanner strategies to deal with metals, full 16-bit dynamic range raw data reconstruction[18–20] and metal reduction artifact (MAR) solutions[21,22] have been investigated. Nevertheless, strong cupping artifacts remain and make the mean HU values of metallic objects strongly dependent on their size.[23,24] Currently, automatic assignment of materials based on a unique extended CT-density calibration curve has been investigated[25] but should be validated with major efforts in clinical practice.[24] Interestingly, some TPS allow extending the CT-density calibration curve to obtain electron or mass density values for high-density materials, whereas others force the user to manually assign the material above a certain HU threshold value. However, to the best of our knowledge, no method can reliably determine the material of these implants.

The type of prosthesis for radiotherapy patients is often unknown, making the determination of the prosthesis material difficult. Information about the manufacturer is often incomplete or missing (shape, size, composition, and effective atomic number) and assumptions about the material have to be made.[25]

To improve the dose calculation accuracy of treatment plans for patients with hip prostheses, we propose a method for determining the type of materials used solely on the basis of CT images.

In this study, we first verified if extending the CT mass density calibration curve was a viable strategy for material determination using titanium (Ti) and stainless steel (SS) rods of two different diameters. We then reviewed and summarized the materials used in the composition of the various parts of the hip prosthesis before classifying them into two groups depending on their physical density: Ti and SS groups. To go beyond the extended CT-density calibration curve strategy, we developed an interpolation function-based algorithm to automatically differentiate and classify the prosthesis material between the Ti and SS groups. Pyramidal reference objects used to create the interpolation functions are presented and analyzed in detail.

The method was applied to different energies and tube filtration for two different reconstruction kernels used in our clinic: a standard body kernel and the electron DirectDensity (eDD) kernel developed by Siemens (Siemens Healthineers, Forchheim, Germany).[26] CT images of reference prosthesis and hip replacement patients treated in our department between 2021 and

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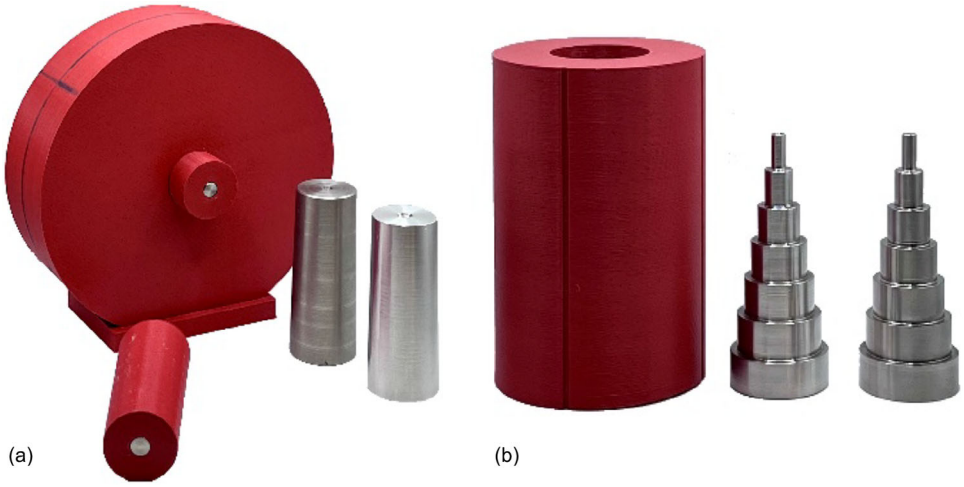


FIGURE 1 (a) 3D printed phantom with small and large diameter titanium and stainless steel rods. (b) The two circular Ti and SS pyramids used as reference objects for the interpolation-based method, along with the 3D printed red cylinder used for holding the pyramids during CT scan acquisitions.

2023 were used to validate our work. Finally, the material and measurements required to adapt this method to other CT scanner units are reviewed, and the limitations and possible improvements of our method are discussed.

2 | MATERIALS AND METHODS

2.1 | CT-Density calibration curve: from air to stainless steel

To study the effect of Ti and SS on CT-density calibration curves, data acquisition was performed on a Siemens SOMATOM go.Open Pro scanner (Siemens Healthineers, Forchheim, Germany) using two phantoms: the CBCT Electron Density Phantom (CIRS, Norfolk, USA)[27] and a homemade phantom (Figure 1a). Acquisition energies were 70, 90, 120, and 140 kV and 100, 120, and 140 kV with the tin (Sn) filter (tube filtration). For each metal, two inserts with diameters 6.5 and 30 mm for Ti (mass density of 4.51 g/cm^3) and 8 and 30 mm for SS (mass density of 7.98 g/cm^3 for 304L) were considered to analyze the impact of the diameter on the mean HU value and HU value profiles. Image reconstructions were performed using two different reconstruction kernels: a standard body kernel (Br44f) and the eDD kernel (Sd40f).[26] The other reconstruction parameters were a 2 mm slice thickness, a 0.58 mm pixel size, an iterative reconstruction algorithm (SAFIRE level 3), and iterative MAR (iMAR) in the hips mode.[28–30] The iterative beam-hardening correction (iBHC) algorithm[31] was turned on only for the Sd40f reconstructions. $CTDI_{vol}$ was chosen to be as high as possible to minimize image noise and thus measurement variations. The mean HU value and its standard deviation (SD) were computed for each insert using a homemade Python 3.10 script. A 5 mm in diameter region of interest (ROI) averaged on five CT slices and centered on the middle of the insert was used to determine the mean and SD values. For the measurements of the insert HU profiles, ImageJ software (https://imagej.net) was used in addition to Python script.

2.2 | Titanium and stainless steel reference objects

To create the interpolation functions presented in this work, two round pyramids made of Ti (4.51 g/cm^3) and SS (7.98 g/cm^3 for 304L) were designed and used as reference objects to develop our method (Figure 1b). The Ti and SS pyramid diameters increase from 5 to 35 mm in 5 mm steps. For CT acquisition, the pyramids were embedded in a 70 mm diameter 3D printed polylactic acid (PLA) cylinder. A PLA infill density of 90%, corresponding to a mass density close to 1.0 g/cm^3, was used to mimic human tissue. The CT scan acquisition and reconstruction parameters were identical to those used in the CT-density calibration curve study.

2.3 | Orthopedic material for hip prostheses

Hip joint prostheses are typically manufactured using metals, ceramics, and plastics. Ti alloys, SS, special high-strength alloys, alumina, zirconia, zirconia toughened alumina (ZTA), and ultra-high-molecular-weight polyethylene (UHMWPE) are predominantly used.[32–37] While stems are usually composed of metals, femoral heads can be both metallic and ceramic, and the acetab-

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TABLE 1 Overview of the most commonly used materials in hip orthopedic surgery.

	Material	Most frequent	Physical density	Advantage	Disadvantage	Use	References
Biological material	Bone	Solid Dense Bone (800 mg/cc HA)—ICRU Cortical Bone	1.53–1.91 g/cm^3	—	—	—	[27]
Metals	Steel	304 L, 316 L	7.93 g/cm^3, 8.03 g/cm^3	High material strength Good ductility	Corrosive Aseptic loosening Inadequate wear resistance	hip stem	[36]
	Ti alloys	Ti6Al4V, Ti6Al7Nb	3.94–4.51 g/cm^3	High biocompatibility Low density Corrosion resistance	Poor tribological properties Low wear resistance Toxic effect of aluminum and vanadium	hip stem acetabular cups of hip	[2, 36]
	CoCrMo alloys	Co28Cr6Mo	7.93–8.3 g/cm^3	High material strength High wear resistance Corrosion resistance	Allergy consideration with nickel chrome and cobalt	femoral head	[36, 37]
Ceramics	Aluminum and zirconium oxides	zirconia-toughened alumina (ZTA)	4.10–4.38 g/cm^3	High hardness Wear resistance Good wettability Good biocompatibility	Brittle High stiffness Low flexibility	femoral head acetabular cups of hip	[32, 36]
Polymeric	Polyethylène	ultra-high molecular weight polyethylene (UHMWPE)	0.930–0.949 g/cm^3	Low density, Biodegradable, Easy fabrication, Flexible	Hard to sterilize Poor tribological properties Absorb water and proteins	acetabular cups of hip liner	[36, 37]

ulum can be made up of metals, ceramics, or polymers. The physical densities of the most frequently used materials in hip orthopedic surgery are presented in Table 1. With the exception of polymeric compounds whose density is close to that of tissues, ceramics and Ti alloys have physical densities around 4.0–5.0 g/cm^3, whereas SS and CoCr alloys have physical densities around 7.9–8.5 g/cm^3. This makes it possible to separate these materials into two categories: the Ti group (4–5 g/cm^3) and the SS group (7.9–8.5 g/cm^3), as already observed by Coolens et al.[23]

In some TPS, such as Eclipse 16.1 treatment planning system (Varian Medical Systems Inc., Palo Alto, California, USA), metals available in the AXB physical material table are Al, Ti alloy, SS, and gold with densities of 2.70, 4.42, 8.00, and 19.32 g/cm^3, respectively. Materials with physical densities higher than 3.0 g/cm^3 must be manually assigned to Ti alloy or SS to calculate the dose.[14]

2.4 | Reference hip prostheses

Figure 2 shows the three hip prostheses used to adjust and test our method. The first prosthesis (a) is a hip stem made of Ti alloy (TiAlNb, Quadra-H from Medacta),[38] the second (b) is a hip stem made of SS (AMI-Stem from Medacta), and the last (c) is a complete hip system (Revitan hip system from Zimmer)[39] made up of different materials. The femoral stem consists of two parts: one proximal and one distal component, both made of Ti6Al7Nb Ti alloy (Protasul−100).[40] A taper connection composed of CoCrMo ensures mechanical coupling. The Revitan system used in this study was mounted with a zirconia-toughened alumina (ZTA) femoral head (Biolox delta from CeramTec).[41] These three prostheses were chosen because their shape and materials are representative of those currently available on the market.

Before CT acquisition, the prostheses were placed in a 105 mm diameter polypropylene (PP) tube filled with rice grains to simulate the morphology of the human leg.[42] The acquisition and reconstruction parameters were identical to those used to obtain the CT-density calibration curve and for the pyramidal reference objects study.

2.5 | Method based on the interpolation functions

This section introduces the method developed in this study. This method can be divided into four parts: (A) object detection and object parameter calculation; (B) object shape validation and object HU calculation; (C) HU calculated from interpolation functions and material attribution; and (D) object overriding and DICOM-RT structure set creation. Figure 3 shows a schematic of

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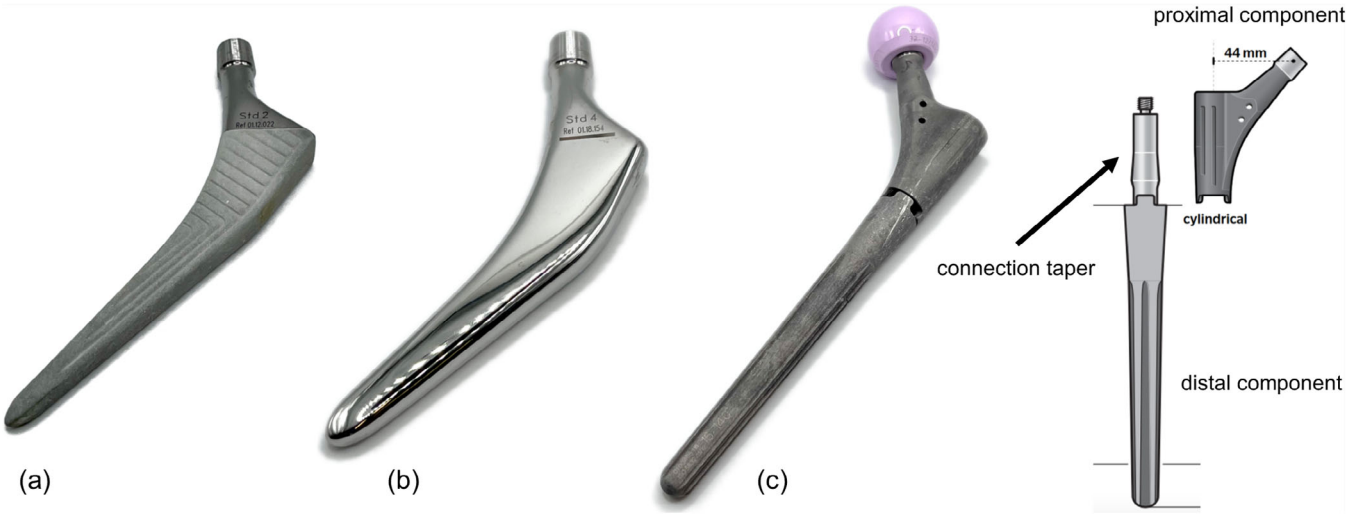


FIGURE 2 The three hip prostheses used to tune and validate the method presented in this work: (a) a TiAlNb stem Quadra-H (Medacta),[38] (b) a stainless steel stem AMI-Stem (Medacta),[38] and (c) the Revitan straight hip system (Zimmer).[39] This system consists of a set of femoral stems made of Ti6Al7Nb linked by a taper connection made of CoCrMo (schematic is adapted from Zimmer). Here, the prosthesis ball head mount on the Revitan system is made of ceramic (Biolox delta from CeramTec).[41]

the strategy used in this method. Although this method is based on certain assumptions (see validation criteria below), it has been optimized and validated to identify objects from 7 to 40 mm in diameter with excellent probability at CT energies from 120 to 140 kV without the Sn filter and 110 to 140 kV with the Sn filter.

2.5.1 | Object detection and parameter calculation

For each CT slice, the objects present in the image are detected using thresholding. To detect objects above bone density (1.82 g/cm^3), threshold values of 1500 and 2500 HU were set for the Sd40f and Br44f kernels, respectively. Each detected object was then resampled with a 256 × 256 matrix using cubic spline interpolation and the lengths of the minor and major axes of the object were calculated using the *skimage.mesure.regionprops()* function present in the scikit-image open-source Python library. The object diameter was approximated as the average of both axes. The HU profiles along the major and minor axes were then obtained by performing an average with the nearest lateral neighbor's pixel (3-pixel average) for each position along the axis. This reduces the HU variations and increases the robustness of the method. Finally, the major and minor HU profile symmetries were calculated for each detected object.

2.5.2 | Object shape validation and HU calculation

Before assigning a material to each detected object in a slice, the object parameters must meet certain criteria defined below to ensure that they are compatible with the reference disk-shaped object (Figure 1b). The farther the shape of the analyzed objects is from the disk shape, the greater is the uncertainty regarding the detected material. To identify the object material with a high probability, the following criteria for object validation must be satisfied:

$$\begin{cases} \text{(i) } 4\text{ mm} < \text{object diameter} < 40\text{ mm} \\ \text{(ii) Symmetry difference for both major and minor ellipse axis profiles} < 20\% \\ \text{(iii) Major and minor axis length difference} < 70\% \\ \text{(iv)} \begin{cases} \text{Difference between major axis length and long side (yellow) box} < 15\% \\ \text{Difference between minor axis length and short side (yellow) box} < 15\% \end{cases} \end{cases}$$

These values were determined empirically by analyzing the results obtained from the hip prostheses. Examples of objects found in the hip prosthesis images

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are shown in Figure 4. Criterion (iv) allows to reject the shape of the object that deviates from the elliptical shape (Figure 4d). With a value of 15%, objects with a shape slightly resembling a rectangle or a doughnut will be accepted. Objects were considered valid for calculating HU and material determination if all of the above criteria were met.

Metal objects do not provide a uniform surface due to cupping artifacts.[20,24] Figure 5a,b show the impact of cupping artifacts on the HU profiles for small and large Ti and SS inserts and for extreme energy values, mainly 70 and 140 kV. The use of a central ROI to determine the mean HU value of very high-density objects should be used with caution, especially when data are acquired at low energy. To produce a robust method, we defined a new HU_{OBJECT} metric based on the major and minor HU profiles [Equation (1)]:

$$\begin{cases} HU_{MAX_major_axis} := (max\,(left\ half\ profile) + max\,(right\ half\ profile))\,/2 & \text{(for the major axis)} \\ HU_{MAX_minor_axis} := (max\,(left\ half\ profile) + max\,(right\ half\ profile))\,/2 & \text{(for the minor axis)} \\ HU_{MAX} := \left(HU_{MAX_major_axis} + HU_{MAX_minor_axis}\right)/2 \\ HU_{CENTER} := \left(HU_{CENTER_major_axis} + HU_{CENTER_minor_axis}\right)/2 \\ \boldsymbol{HU_{OBJECT}} := (\boldsymbol{HU_{MAX}} + \boldsymbol{HU_{CENTER}})\,/2 \end{cases} \quad (1)$$

The HU_{OBJECT} metric provides a reliable value for convex, flat, or concave profiles, as illustrated in Figure 5c. In parallel, using predefined interpolation functions (see below), two additional metrics HU_{OBJECT_Ti}, HU_{OBJECT_SS} are computed for the same diameter and CT parameters as the detected object.

2.5.3 | HU calculated from the interpolation functions $\mathscr{F}_x$ and material attribution

To allow the method to determine materials different from the reference objects, two interpolation functions, called $\mathscr{F}_x$ where x corresponds to Ti or SS, were used. These functions $\mathscr{F}_x$, were built using a bivariable cubic spline interpolation to calculate the metric HU_{OBJECT} [Equation (1)] for each diameter of the two reference objects and by varying the CT acquisition and reconstruction parameters. Each $\mathscr{F}_x$ depends on the object diameter, reconstruction kernel, tube voltage, and Sn filter. Figure 6 shows the $\mathscr{F}_x$ functions of each metal for the two reconstruction kernels with or without the Sn filter.

Then, the percentage differences between HU_{OBJECT} and the two metrics HU_{OBJECT_Ti} and HU_{OBJECT_SS} were computed. The metal providing the smallest difference was considered to be the most likely material constituting the detected object. This material was then assigned to the detected object.

TABLE 2 Summary of the information obtained from surgery reports for the 18 patients included in this study.

No. of patients	18
No. of implants	21 (19 prosthesis, 2 nails)
Stems/nails	21 (12 Ti, 7 SS, 2 n.s.)
Heads	19 (9 ceramic, 5 SS, 3 CrCo, 2 n.s.)
Acetab. cups	19 (16 Ti, 2 SS,1 n.s.)
Data sets	40 (23 with Sd40f, 17 with Br44f)

n.s = not specified.

2.5.4 | Object overriding and DICOM-RT structure set creation

A new set of images was created by overriding the HU values of the validated objects with arbitrarily defined Ti and SS HU values. Then, a DICOM-RT structure set was generated by thresholding these HU values, using the open-source platform 3D Slicer (www.slicer.org) with a Python script. Finally, the DICOM-RT structure set was imported into Eclipse, and the Ti and SS contours were completed when necessary.

To simplify the user's task, the entire process has been automated: when a hip prosthesis is visible on the CT images, the user sends the entire CT data set to a DICOM node and a DICOM-RT structure set is automatically generated.

2.6 | Clinical validation

Table 2 summarizes the forty data sets from 18 different patients with a single or dual hip prosthesis that were used to validate the identification method proposed in this study. Pelvic images were obtained from patients presenting with prostate cancer or metastatic disease treated in our department between 2021 and 2023. In our clinic, the Sd40f kernel is used systematically

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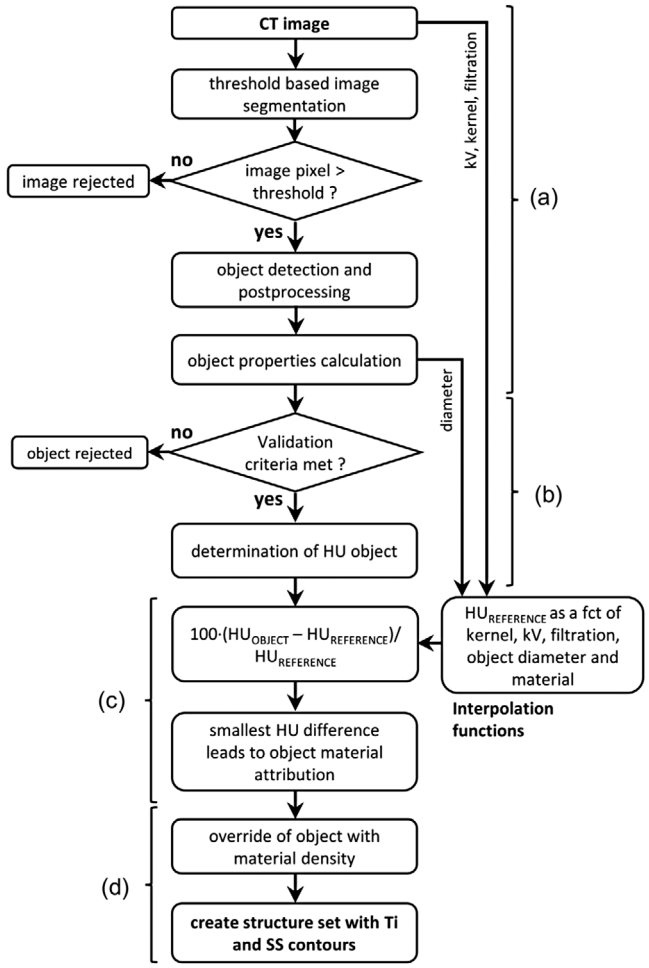


FIGURE 3 Schematic representation of the strategy used to differentiate between Ti and SS material groups in hip prostheses. The method can be divided into four parts: (a) object detection and object parameter calculation; (b) object shape validation and object HU calculation; (c) HU calculated from interpolation functions and material attribution; and (d) object overriding and DICOM-RT structure set creation.

for dosimetry calculations, whereas the Br44f kernel is used only when necessary for contour delineation. Consequently, 23 and 17 CT image reconstructions were obtained using Sd40f and Br44f, respectively. To be included in the study, the operative report for each implanted prosthesis in our hospital had to be available. All patients provided signed informed consent to participate in this study.

2.7 | Uncertainties and sensitivity analysis

The uncertainty of the distance measurement was estimated to be 2 pixels (before interpolation), while the uncertainty of the HU measurement was estimated to be 5% (this value is conservative as the HU profiles are already an average along the object axis

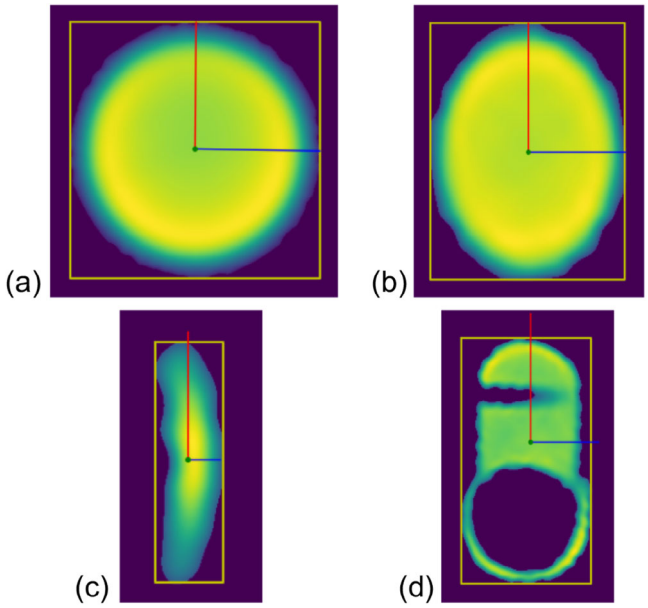


FIGURE 4 Examples of objects found in hip prosthesis images. Objects (a) and (b) meet all validation criteria, whereas object (c) does not meet criteria (ii) and (iii), and object (d) does not meet criteria (ii) and (iv).

with the nearest neighbors). The uncertainties were then propagated to produce uncertainty ranges on the interpolation functions and uncertainty bars on the object diameter and HU_{OBJECT}. Sensitivity analyses were performed for each prosthesis to evaluate the robustness of the method and to determine its validity range.

3 | RESULTS

3.1 | CT-Density calibration curves extended for metallic objects

Figure 7 shows the extended HU calibration curves with Ti and SS materials for the Br44f and Sd40f (eDD) reconstruction kernels. The results obtained from air to bone materials for both kernels agree with those reported in the literature.[26,43,44] For Ti and SS inserts, the HU values are directly dependent on the CT energy spectrum (choice of kV and Sn filter) for Br44f and Sd40f.[45,46] Indeed, for materials above bone density, eDD is unable to produce a unique calibration curve. Moreover, the mean HU values of the high-density inserts depend on their diameter due to cupping artifacts, as shown in Table 3.

For the Ti inserts, the mean HU value decreased from 16 842 ± 1604 to 6721 ± 117 HU at 70 kV and from 7783 ± 439 HU to 5805 ± 23 HU at 140 kV when the insert diameter increased from 6.5 to 30 mm using the Br44f kernel. This corresponds to a reduction of

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TABLE 3 Mean HU values and standard deviation for the different inserts of the CIRS and 3D printed phantoms.

	Br44f								Sd40f					
	70 kV	90 kV	120 kV	140 kV	Sn100 kV	Sn120 kV	Sn140 kV	70 kV	90 kV	120 kV	140 kV	Sn100 kV	Sn120 kV	Sn140 kV
Air 0 g/cm^3	−1003 ± 9	−1003 ± 5	−1002 ± 3	−1001 ± 3	−1002 ± 7	−1002 ± 5	−1002 ± 3	−1005 ± 5	−1004 ± 3	−1003 ± 2	−1002 ± 2	−1002 ± 3	−1002 ± 3	−1002 ± 3
Lung inhale 0.2 g/cm^3	−851 ± 13	−850 ± 8	−846 ± 9	−845 ± 8	−849 ± 12	−847 ± 9	−847 ± 6	−853 ± 9	−850 ± 7	−846 ± 7	−845 ± 7	−847 ± 8	−845 ± 7	−845 ± 7
Lung exhale 0.5 g/cm^3	−497 ± 14	−499 ± 8	−499 ± 6	−500 ± 6	−501 ± 9	−500 ± 8	−500 ± 5	−497 ± 8	−499 ± 6	−498 ± 4	−499 ± 4	−499 ± 5	−499 ± 5	−500 ± 4
Adipose 0.96 g/cm^3	−44 ± 13	−38 ± 7	−34 ± 6	−31 ± 5	−33 ± 10	−30 ± 8	−30 ± 5	−49 ± 8	−42 ± 4	−36 ± 3	−34 ± 2	−34 ± 6	−30 ± 4	−28 ± 3
Breast 0.99 g/cm^3	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12	−83 ± 12
H_2O 1.00 g/cm^3	−29 ± 17	−26 ± 11	−20 ± 7	−20 ± 7	−6 ± 12	−7 ± 6	−7 ± 4	28 ± 9	9 ± 5	4 ± 3	−1 ± 3	1 ± 7	−2 ± 3	−1 ± 5
Muscle 1.06 g/cm^3	54 ± 16	48 ± 7	45 ± 6	45 ± 4	46 ± 9	45 ± 6	45 ± 4	53 ± 9	47 ± 4	44 ± 4	44 ± 3	45 ± 5	44 ± 4	44 ± 3
Liver 1.07 g/cm^3	65 ± 13	58 ± 7	55 ± 5	54 ± 4	55 ± 10	53 ± 5	51 ± 4	63 ± 7	57 ± 4	54 ± 3	53 ± 2	53 ± 5	53 ± 3	51 ± 4
Bone 200 mg/cc 1.16 g/cm^3	380 ± 14	289 ± 8	232 ± 6	211 ± 5	219 ± 10	189 ± 7	164 ± 5	124 ± 5	121 ± 3	121 ± 3	121 ± 3	127 ± 3	119 ± 3	123 ± 3
Bone 800 mg/cc 1.53 g/cm^3	1470 ± 20	1135 ± 12	904 ± 8	816 ± 7	868 ± 12	754 ± 9	673 ± 7	541 ± 9	511 ± 9	493 ± 5	485 ± 3	508 ± 4	474 ± 3	486 ± 3
Bone 1250 mg/cc 1.82 g/cm^3	2254 ± 22	1735 ± 14	1372 ± 10	1238 ± 8	1331 ± 14	1157 ± 11	1087 ± 9	954 ± 8	866 ± 5	817 ± 3	797 ± 2	799 ± 4	745 ± 4	763 ± 3
Ti (large) 4.51 g/cm^3	6721 ± 117	7730 ± 94	6510 ± 35	5805 ± 23	7011 ± 50	6003 ± 31	5389 ± 16	5053 ± 77	6595 ± 56	5760 ± 23	5193 ± 14	6209 ± 23	5325 ± 16	4832 ± 12
Ti (small) 4.51 g/cm^3	16842 ± 1604	12379 ± 794	9024 ± 526	7783 ± 439	9211 ± 609	7380 ± 455	5842 ± 304	12829 ± 1827	9513 ± 1039	7029 ± 710	6115 ± 599	7286 ± 791	5818 ± 609	5115 ± 514
SS (large) 8.10 g/cm^3	5470 ± 224	6110 ± 187	7166 ± 223	7834 ± 199	6368 ± 238	7529 ± 221	8524 ± 220	3803 ± 194	5027 ± 173	6576 ± 206	7412 ± 177	5706 ± 198	7003 ± 181	7894 ± 152
SS (small) 8.10 g/cm^3	26670 ± 2706	25028 ± 1341	19192 ± 636	16586 ± 487	21938 ± 645	17579 ± 379	14957 ± 198	23986 ± 1850	22818 ± 884	17596 ± 424	15246 ± 331	20067 ± 354	16059 ± 204	13809 ± 147

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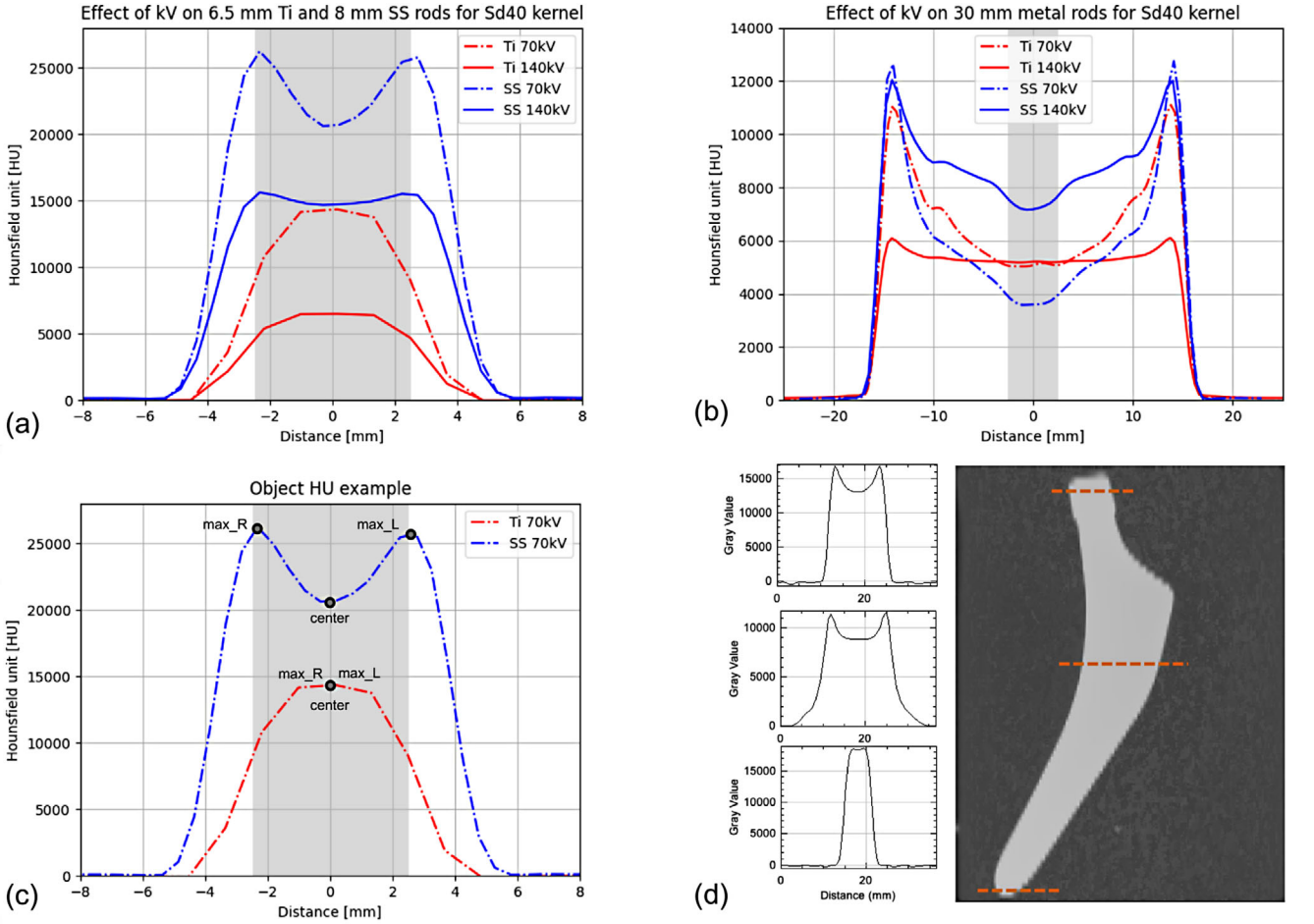


FIGURE 5 HU rod profiles for the Sd40f reconstruction kernel when image acquisition was performed at 70 and 140 kV. (a) HU profiles obtained for the small Ti and SS rods. (b) HU profiles obtained for the large Ti and SS rods. (c) Definition of the metric introduced in this work to measure HU when object profiles are inhomogeneous: HU_{OBJECT} = (average(max_R, max_L) + center)/2 (Equation (1)). (d) Example of HU profiles as a function of position for the SS hip prosthesis shown in Figure 2. The image was acquired at 120 kV and reconstruction was performed using the Sd40f kernel. The gray region corresponds to the ROI of 5 mm in diameter used to determine the HU mean and SD values of the inserts.

60 ± 11% and 25 ± 22% at 70 and 140 kV, respectively. For the SS inserts, the impact of the object diameter is increased due to the higher density and stronger cupping artifacts. When the diameter varies from 8 to 30 mm, reductions of 79 ± 6% and 53 ± 5% are obtained at 70 and 140 kV with Br44f, respectively. Figure 7 shows that for Br44f at 120 kV, the small Ti rod (9024 ± 526 HU) has a higher average HU value than the large SS rod (7166 ± 223 HU). Thus, a large SS object with a diameter of 30 mm results in an object with a density of approximately 3.8 g/cm^3 and will be identified as Ti. These results show that a linear extrapolation method based on small rods can lead to incorrect material assignments for larger objects. The same trend was observed for Sd40f (Figure 7). Although the chosen ROI diameter of 5 mm is small, it has a measurable effect on the HU due to cupping artifacts that create non-uniform object profiles (Figure 5).

3.2 | HU profiles of the metallic objects

To analyze the effect of cupping artifacts in detail, the profiles of the metal objects were plotted for the two most distant energies, namely 70 and 140 kV. Figure 5a,b show the impact of energy and insert diameter on the profiles of both materials when images are reconstructed with Sd40f. For low-energy and high-density objects with large diameters, beam hardening is maximized and a strong underestimation of the object density at its center is obtained, as described in.[17,47] In this work, the phenomenon is maximized with the Ti insert of 30 mm in diameter. The difference in HU between the object center and its edges (*horns*) is 6071 ± 806 HU at 70 kV and decreases to 920 ± 563 HU at 140 kV. For the SS insert of the same diameter, 9168 ± 817 and 4855 ± 959 HU differences were obtained at 70 and 140 kV, respectively. Due to their

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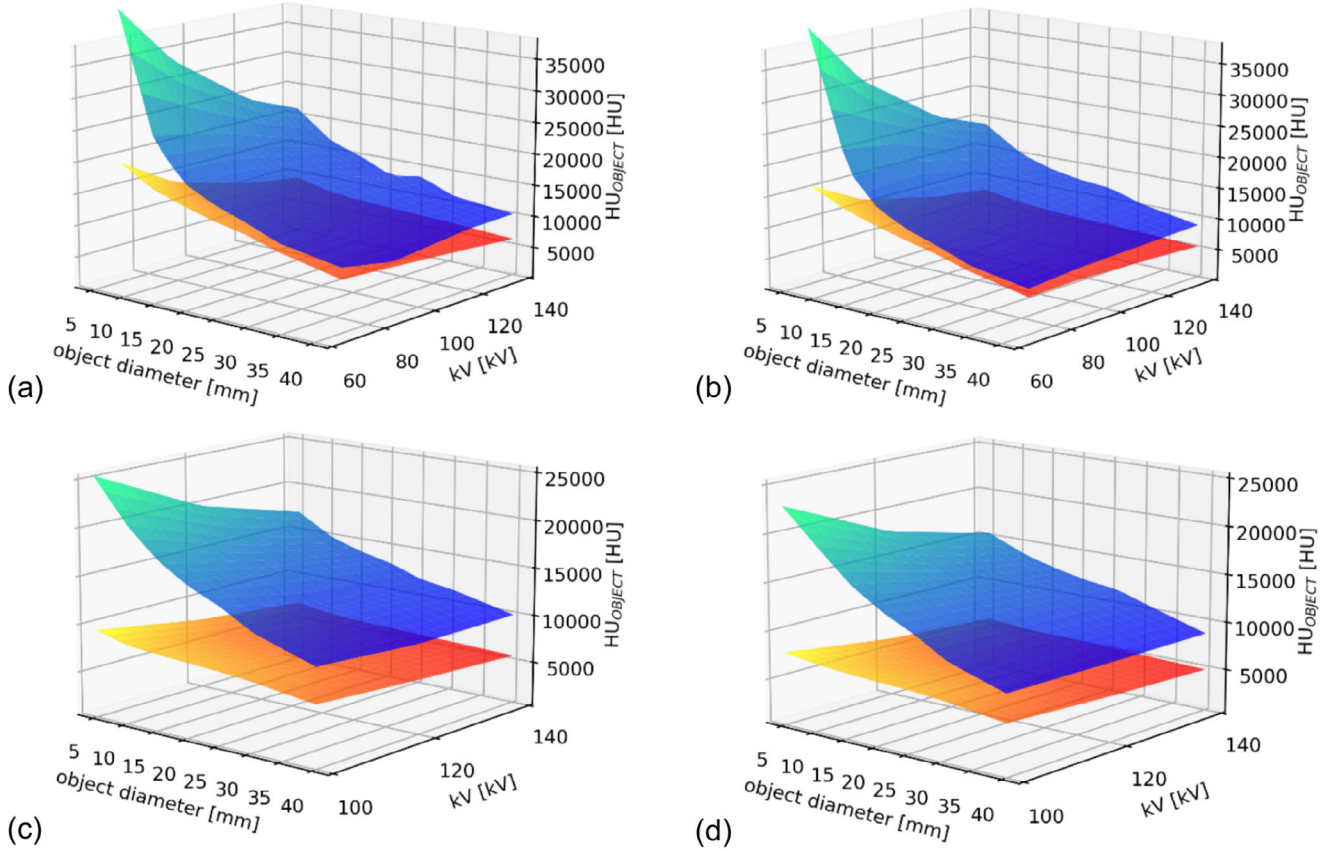


FIGURE 6 3D plots of the interpolation functions $\mathscr{F}_{x,}$ (where *x* equals to Ti or SS) when the reconstruction kernel and Sn filter are: (a) Br44f without Sn filter, (b) Sd40f without Sn filter, (c) Br44f with Sn filter and (d) Sd40f with Sn filter. For each plot, red-orange and blue-green surfaces correspond to $\mathscr{F}_{Ti}$ and $\mathscr{F}_{SS}$, respectively.

smaller diameters, the 6.5-mm Ti and 8-mm SS inserts are less impacted. Figure 5d shows the SS prosthesis (Figure 2b) when imaged at 120 kV and data are reconstructed with Sd40f. As before for the reference objects, HU values at the center of the prosthesis are different along the longitudinal axis of the prosthesis because of cupping artifacts. Note that the Sd40f reconstruction kernel is shown, but a similar behavior is obtained using the Br44f kernel.

3.3 | Impact of the reconstruction kernels on the HU profiles of reference objects

The method based on the interpolation functions relies on the diameter and HU measurements of Ti and SS reference objects (Figure 1b). Figure 8 shows the transverse and longitudinal profiles of the three pyramids for the Br44f and Sd40f reconstruction kernels acquired at 120 kV. This shows that eDD is slightly less affected by metal artifacts than the standard kernel reconstruction. This can probably be explained by eDD automatically turning on the iBHC algorithm. However, for both metals, longitudinal and transverse HU profiles vary too widely to be considered independent of object diameter. For Br44f, a decrease in the HU values on the longitudinal axis of the object is observed for the Ti and SS pyramids. Maximum relative differences are $-42 \pm 17\%$ and $-73 \pm 5\%$ for Ti and SS, respectively, when the diameter increases from the smallest to the largest. For Sd40f, the same observation can be made.

3.4 | Validation on the reference hip prostheses

Figure 9 shows the three hip prostheses (the Ti, SS, and Revitan prosthesis) presented in Figure 2. For each prosthesis, the real image (left), the 3D reconstruction (middle), and the results obtained using our method (right) are presented. The yellow and blue parts of the prostheses were identified as Ti and SS, respectively. Figure 9c shows that the Biolox delta head made of ceramic and the CoCrMo taper connection of the Revitan system are detected as Ti and SS. Parts of the prostheses without a defined color correspond to CT

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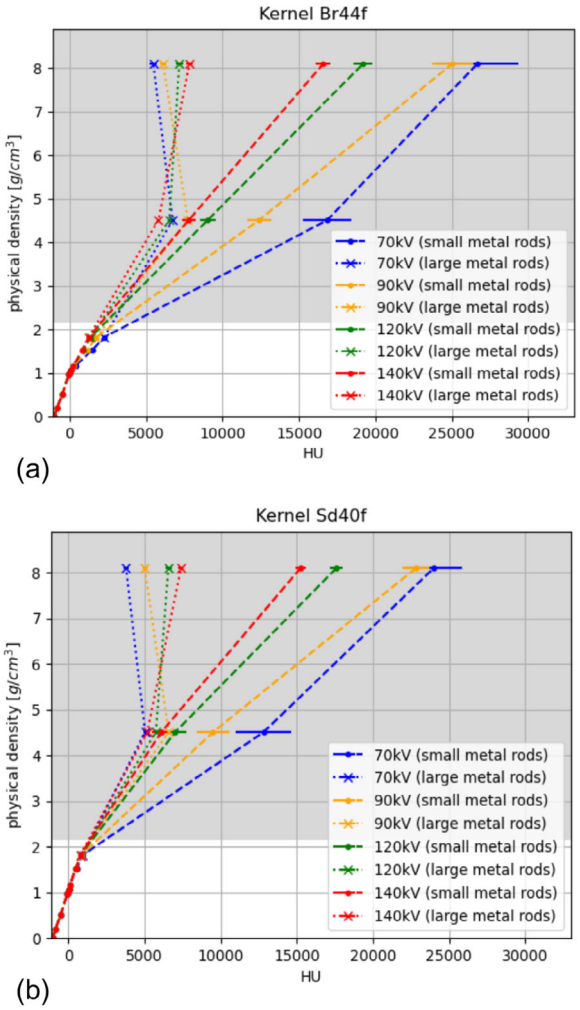


FIGURE 7 Calibration curves extended for Ti and SS materials for the two different reconstruction kernels considered in this work: (a) Br44f and (b) Sd40f (electron DirectDensity). Two different rod diameters were considered for Ti (6.5 and 30 mm) and SS (8 and 30 mm). Measurements were performed using a 5 mm diameter ROI placed at the center of the object.

slices in which the objects have not met the object validation criteria (Figure 9a,c).

Figure 10 shows the results obtained for the Revitan system as a function of kV and the Sn filter. Material differentiation results strongly depend on the CT spectrum and weakly on the reconstruction kernels. For energies below 120 kV without the Sn filter and 100 kV with the Sn filter, detectability is reduced for both kernels.

3.5 | Validation on clinical data

The detection and material differentiation results obtained from hip replacements of the 18 patients included in this study are shown in Tables 4 and 5. For each prosthesis part, the number of objects detected and the percentage of objects correctly identified are given. As shown in Figure 11, very few stem parts were rejected due to inappropriate geometry. Prosthetic heads and acetabular cup parts were more challenging to detect and differentiate due to their complex shapes.

For the prosthesis heads, at least part of the head was detected. This was sufficient for correct material differentiation in all but one patient (patient 11 had two hip prostheses and was scanned at 110 kV). The acetabular cups were rarely fully detected. Most of the time, only the top parts of the cups were visible, as shown in Figure 11a,c–e. However, despite the low number of cup objects detected, superimposing the identification results with a 3D reconstruction of the prosthesis allowed us to differentiate between the head and the cup and correctly attribute the material. This method was also able to detect and correctly identify screws in patients, as shown in Figure 11b.

For the validated energy range of our method (120–140 kV without the Sn filter and 110–140 kV with the Sn filter), Table 6 summarizes the results shown in Tables 4 and 5. For the reconstruction with the Sd40f kernel, 1352 out of 1356 objects belonging to the stems were correctly identified, leading to an identification rate of 99.7%. For the heads and cups, 144 out of 148 and 67 out of 69 objects were correctly identified allowing identification rates of 97.3% and 97.1%, respectively. These high success rates allow the correct differentiation between Ti and SS for 20 out of 20 (100%) for the stems, 18 out of 18 for the heads (100%), and 19 out of 20 cups (95.0%). The corresponding detection rates were 100%, 94.7%, and 100%, respectively. The results obtained using the Br44f kernel were similar as shown in Tables 5 and 6.

Figure 12 shows the results of the sensitivity analysis for the patients 1 and 18 presented in Figure 11 and for all patient hip prostheses imaged at the lowest validated energies (120 kV without the Sn filter and 110 kV with the Sn filter). The equiprobability curve corresponds to the boundary between Ti and SS. An object on this boundary is equally likely to be detected as Ti or SS. The red circles represent the objects whose material has been falsely identified. Uncertainties in the object diameter have little effect on the robustness of the proposed method. Indeed, a diameter shift does not lead to a significant error in HU_{OBJECT} because the slopes of the Ti and SS reference curves are not sufficiently large under the validation conditions of our model (diameters from 7 to 40 mm). The effect of uncertainty on the calculation of HU_{OBJECT} only affects objects close to the equiprobability curve. This mainly concerns large diameter objects and objects with shapes far from the reference object shape (disk shape). Sensitivity analysis confirms that object material differentiation is highly dependent on object shape. As shown in Figure 12, prosthesis head and cup objects are furthest from the Ti and SS reference curves and thus the most difficult to differentiate. However, the chosen object validation criteria allow for a

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TABLE 4 Detection and material differentiation results obtained using the method presented in this study. Patient images were reconstructed using the Sd40f reconstruction kernel.

Patient	kV	Kernel	Filter Sn	Part	Material	No. of objects detected as	Correct identification (%)
1	140 kV	Sd40f	no	stem	Ti + SS (Revitan 140 × 22 mm Zimmer)	Ti: 71, SS: 29	yes (100%)
				head	SS (Protasul 36 Zimmer)	SS: 16	yes (93.8%)
				acetab. cup	Ti (Fitmore 60 Zimmer)	Ti (top): 3	yes (100%)
2	140 kV	Sd40f	no	stem	SS (MS-30 6 Zimmer)	SS: 54	yes (98.1%)
				head	SS (Protasul 36 Zimmer)	SS: 18	yes (94.4%)
				acetab. cup	Ti (Fitmore 54 Zimmer)	Ti (top): 2	yes (100%)
3	120 kV	Sd40f	no	stem	TiAl6V4 (Corail std 12 DePuy Synthes)	Ti: 69	yes (100%)
				head	ZTA (Biolox delta 36 + 5 CeramTec)	Ti: 14	yes (100%)
				acetab. cup	*not specified*	*Ti (top): 1*	—
4	120 kV	Sd40f	no	stem	TiAl6V4 (Corail std 14 DePuy Synthes)	Ti: 89	yes (100%)
				head	*not specified*	*Ti: 4*	—
				acetab. cup	SS (Bi-Mentum 53 DePuy Synthes)	SS (top): 2	yes (100.0%)
5	120 kV	Sd40f	no	nail	Ti alloy (LFN 380, 12)	Ti: 175	yes (100%)
6	90 kV	Sd40f	no	nail	Ti alloy (TFNA 200, 11)	Ti: 91	yes (98.9%)
7	120 kV	Sd40f	no	stem	SS (MS-30 10 Zimmer)	SS: 65	yes (98.5%)
				head	SS (Protasul 36 Zimmer)	SS: 11	yes (90.9%)
				acetab. cup	Ti (Fitmore 56 Zimmer)	Ti (top): 1	yes (100%)
8	120 kV	Sd40f	no	stem	Ti6Al4V (twinSys uncemented 9 Mathys)	Ti: 67	yes (100%)
				head	*not specified*	*SS: 1*	—
				acetab. cup	Ti6Al7Nb (seleXys 48 Mathys)	Ti: 5	yes (80.0%)
9	120 kV	Sd40f	yes	stem	TiAl6V4 (Corail std 11 DePuy Synthes)	Ti: 70	yes (100%)
				head	ZTA (Biolox delta 36 + 5 CeramTec)	Ti: 4	yes (100%)
				acetab. cup	Ti (Pinnacle 56 DePuy Synthes)	Ti: 6	yes (100%)
10	110 kV	Sd40f	yes	stem	SS (AMIStem cemented)	SS: 54	yes (98.1%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 2	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 50 Medacta)	Ti (top): 4	yes (100%)
11	110 kV	Sd40f	yes	stem	*not specified*	*Ti: 68*	—
				head	CrCo 28 + 8.5	not detected	—
				acetab. cup	SS (Bi-Mentum 55 DePuy Synthes)	Ti (top): 1	no (0%)
				stem	TiAl6V4 (Corail std 12 DePuy Synthes)	Ti: 74	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 13	yes (100%)
				acetab. cup	Ti (Pinnacle 58 DePuy Synthes)	Ti: 6	yes (100%)
	140 kV	Sd40f	yes	stem	*not specified*	*Ti: 66*	—
				head	CrCo 28 + 8.5	SS: 9	yes (100%)
				acetab. cup	SS (Bi-Mentum 55 DePuy Synthes)	SS (top): 3	yes (100%)
				stem	TiAl6V4 (Corail std 12 DePuy Synthes)	Ti: 76	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 13	yes (100%)
				acetab. cup	Ti (Pinnacle 58 DePuy Synthes)	Ti (top): 3	yes (100%)
12	110 kV	Sd40f	yes	stem	SS (MS-30 10 Zimmer)	SS: 66	yes (100%)
				head	CrCo 36	SS: 2	yes (100%)
				acetab. cup	Ti (Fitmore 52 Zimmer)	Ti (top): 4	yes (100%)

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TABLE 4 (Continued)

Patient	kV	Kernel	Filter Sn	Part	Material	No. of objects detected as	Correct identification (%)
13	110 kV	Sd40f	yes	stem	Ti (AMIStem std uncemented 0 Medacta)	Ti: 47	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 3	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 50 Medacta)	Ti (top): 4	yes (100%)
	110 kV	Sd40f	yes	stem	Ti (AMIStem std uncemented 2 Medacta)	Ti: 50	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 3	yes (100%)
				acetab. cup	Ti (Versafit® Trio CC 54 Medacta)	Ti: 4	yes (100%)
14	110 kV	Sd40f	yes	stem	*not specified*	*Ti: 57*	—
				head	ZTA (Biolox delta 32 CeramTec)	Ti: 11	yes (100%)
				acetab. cup	Ti (MPact 60 Medacta)	Ti (top): 1	yes (100%)
15	120 kV	Sd40f	no	stem	SS (MS-30 14 Zimmer)	SS: 71	yes (98.6%)
				head	CrCo M/36	SS: 11	yes (90.9%)
				acetab. cup	Ti (Fitmore 64 Zimmer)	Ti (top): 2	yes (100%)
16	120 kV	Sd40f	no	stem	TiNb alloy (Quadra 4 Medacta)	Ti: 60	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 2	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 58 Medacta)	Ti:7	yes (100%)
	120 kV	Sd40f	no	stem	TiNb alloy (Quadra 4 Medacta)	Ti: 65	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 1	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 64 Medacta)	Ti: 6	yes (100%)
17	110 kV	Sd40f	yes	stem	SS (MS-30 8 Zimmer)	SS: 60	yes (100%)
				head	SS (Protasul 36 Zimmer)	SS (top): 1	yes (100%)
				acetab. cup	Ti (Fitmore 54 Zimmer)	Ti (top): 2	yes (100%)
18	140 kV	Sd40f	yes	stem	SS (MS-30 6 Zimmer)	SS: 54	yes (100%)
				head	SS (Protasul 32 Zimmer)	SS: 14	yes (100%)
				acetab. cup	Ti (Fitmore 50 Zimmer)	Ti (top): 3	yes (100%)

very low rate of identification errors and to correctly discriminate between Ti and SS hip prosthesis parts even with few objects detected. The energy-dependent shifts in HU_{OBJECT} between stem objects and the Ti reference curve (Figure 12c,d) can be explained by the differences in Ti alloys and pure Ti attenuation coefficients. Based on the NIST data, the difference between Ti6Al7Nb and Ti at 80 kV photon energy is approximately 22%. Attenuation coefficient differences also exist for Co28Cr6Mo and SS (304L).

4 | DISCUSSION

Recent literature reviews on hip replacements,[33,36] principally focusing on the most relevant research published between 1950 and 2020, show that hip prosthesis materials commonly used include SS, CoCr and Ti alloys, ceramics, polyethylene, and poly(methyl methacrylate) (PMMA). While bearing surfaces are critical for the longevity of hip prostheses, from a dosimetry point of view, stem, head, and acetabular cup prosthesis parts made of SS, CoCr, Ti alloys, and ceramics are the most critical.

Prosthesis stems commercially available are typically made of metal SS, Ti, and CoCr alloys.[38,40,48–53] Although SS remains widely used, it is gradually being supplanted by other alloys, particularly titanium alloys. In contrast to the stems, the heads can also be made of ceramic material.[41] The standards used are CoCr and Biolox delta ceramic.[48,50] However, the trend is an increase in ceramic head usage and a decrease in CoCr usage, as explained by concerns about corrosion.[5] The most complex part with various designs is the acetabular cup. Highly biocompatible materials with excellent corrosion resistance are used. Moreover, to facilitate tissue growth and vascularization, a porous coating on the outer surface is added to the fixing holes. The main materials used for the cups are Ti and SS.[38,40,49–53]

As already observed by Coolens et al.[23] and shown in Soliman et al.,[37] hip prosthesis materials can be separated into two groups (Ti group around 4.0–5.0 g/cm^3 and SS group around 7.5–8.5 g/cm^3) according to their physical density (or mass density), as summarized in Table 1. This particularity allows us to use only Ti and SS reference objects to develop the proposed method.

The origin of this AAPM TG-63 recommendation lies in the fact that, in the past, calculation algorithms did

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TABLE 5 Detection and material differentiation results obtained using the method presented in this study.

Patient	kV	Kernel	Filter Sn	Part	Material	No. of objects detected as	Correct identification (%)
1	140 kV	Br44f	no	stem	Ti + SS (Revitan 140 × 22 mm Zimmer)	Ti: 71, SS: 29	yes (100%)
				head	SS (Protasul 36 Zimmer)	SS: 16	yes (93.8%)
				acetab. cup	Ti (Fitmore 60 Zimmer)	Ti (top): 3	yes (100%)
3	120 kV	Br44f	no	stem	TiAl6V4 (Corail std 12 DePuy Synthes)	Ti: 71	yes (100%)
				head	ZTA (Biolox delta 36 + 5 CeramTec)	Ti: 16	yes (100%)
				acetab. cup	*not specified*	*Ti: 5*	—
4	120 kV	Br44f	no	stem	TiAl6V4 (Corail std 14 DePuy Synthes)	Ti: 90	yes (100%)
				head	*not specified*	*Ti: 10*	—
				acetab. cup	SS (Bi-Mentum 53 DePuy Synthes)	Ti/SS (top): 4	no (25.0%)
6	90 kV	Br44f	no	nail	Ti alloy (TFNA 200, 11)	Ti: 85	yes (98.8%)
9	120 kV	Br44f	yes	stem	TiAl6V4 (Corail std 11 DePuy Synthes)	Ti: 71	yes (100%)
				head	ZTA (Biolox delta 36 + 5 CeramTec)	Ti: 2	yes (100%)
				acetab. cup	Ti (Pinnacle 56 DePuy Synthes)	Ti: 7	yes (100%)
10	110 kV	Br44f	yes	stem	SS (AMIStem cemented)	SS: 55	yes (94.5%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 3	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 50 Medacta)	Ti (top): 2	yes (100%)
11	110 kV	Br44f	yes	stem	*not specified*	*Ti: 65*	—
				head	CrCo 28 + 8.5	not detected	—
				acetab. cup	SS (Bi-Mentum 55 DePuy Synthes)	Ti (top): 1	yes (100%)
				stem	TiAl6V4 (Corail std 12 DePuy Synthes)	Ti: 75	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 12	yes (100%)
				acetab. cup	Ti (Pinnacle® 58 DePuy Synthes)	Ti (top): 4	yes (100%)
	140 kV	Br44f	yes	stem	*not specified*	*Ti: 57*	—
				head	CrCo 28 + 8.5	SS: 9	yes (100%)
				acetab. cup	SS (Bi-Mentum 55 DePuy Synthes)	SS (top): 1	yes (100%)
				stem	TiAl6V4 (Corail std 12 DePuy Synthes)	Ti: 78	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 12	yes (100%)
				acetab. cup	Ti (Pinnacle 58 DePuy Synthes)	Ti (top): 3	yes (100%)
12	110 kV	Br44f	yes	stem	SS (MS-30 10 Zimmer)	SS: 62	yes (100%)
				head	CrCo 36	SS: 2	yes (100%)
				acetab. cup	Ti (Fitmore 52 Zimmer)	Ti (top): 4	yes (100%)
13	110 kV	Br44f	yes	stem	Ti (AMIStem std uncemented 0 Medacta)	Ti: 47	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 2	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 50 Medacta)	Ti (top): 4	yes (100%)
	110 kV	Br44f	yes	stem	Ti (AMIStem std uncemented 2 Medacta)	Ti: 52	yes (100%)
				head	ZTA (Biolox delta 36 CeramTec)	Ti: 3	yes (100%)
				acetab. cup	Ti (Versafit Trio CC 54 Medacta)	Ti (top): 4	yes (100%)
14	110 kV	Br44f	yes	stem	*not specified*	*Ti: 56*	—
				head	ZTA (Biolox delta 32 CeramTec)	Ti: 13	yes (100%)
				acetab. cup	Ti (MPact 60 Medacta)	not detected	—
15	120 kV	Br44f	no	stem	SS (MS-30 14 Zimmer)	SS: 69	yes (98.6%)
				head	CrCo M/36	SS: 10	yes (70.0%)
				acetab. cup	Ti (Fitmore 64 Zimmer)	Ti (top): 3	yes (66.7%)

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TABLE 5 (Continued)

Patient	kV	Kernel	Filter Sn	Part	Material	No. of objects detected as	Correct identification (%)
17	110 kV	Br44f	yes	stem	SS (MS-30 8 Zimmer)	SS: 54	yes (100%)
				head	SS (Protasul 36 Zimmer)	SS: 7	no (28.6%)
				acetab. cup	Ti (Fitmore 54 Zimmer)	Ti (top): 1	yes (100%)
18	140 kV	Br44f	yes	stem	SS (MS-30 6 Zimmer)	SS: 52	yes (100%)
				head	SS (Protasul 32 Zimmer)	SS: 15	yes (93.3%)
				acetab. cup	Ti (Fitmore 50 Zimmer)	Ti (top): 3	yes (100%)

Patient images were reconstructed using the Br44f reconstruction kernel.

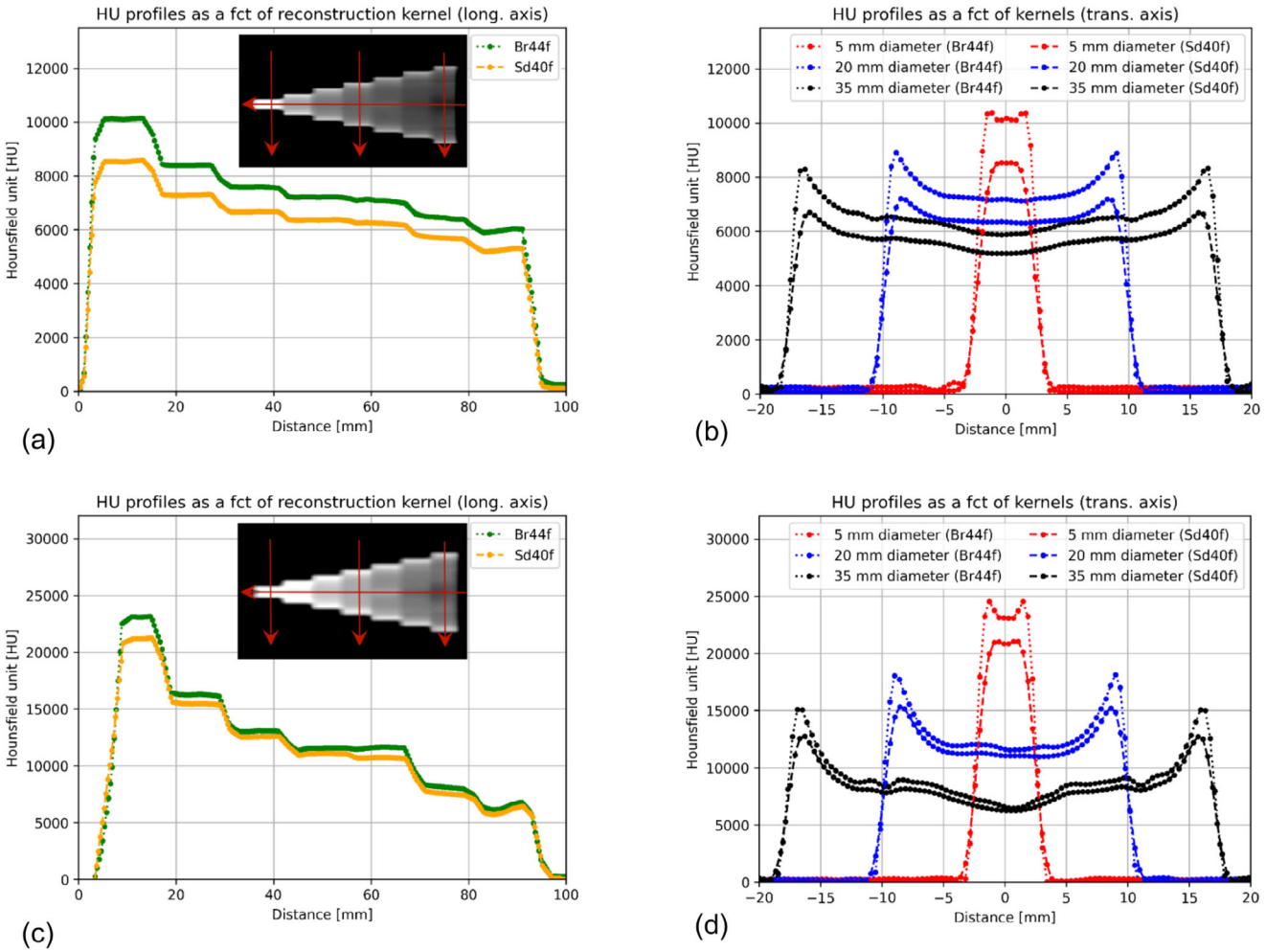


FIGURE 8 HU profiles of the two pyramids (reference objects) when imaged at 120 kV: (a) and (c) are the longitudinal profiles for the Ti and SS pyramids, respectively. (b) and (d) are the transverse profiles evaluated at the large, medium and small diameters of the Ti and SS pyramids, respectively.

not correctly model changes in charged particle production and photon scattering due to changes in material composition. CT images suffered from strong artifacts, especially in the case of bilateral prostheses.[54]

Catli et al. compared the pencil beam convolution (PBC) method implemented in Eclipse 8.6 TPS with the MC method and ionization chamber measurements when treatment is delivered with a unique treatment field through a hip prosthesis.[55] They observed that PBC caused an overdose of 9% and 20.0% for Ti alloy and 4% and 17.2% for SS at 6 and 18 MV, respectively, at a distance of 0.5 cm behind the hip prostheses.

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FIGURE 9 Results obtained for the three hip prostheses (120 kV, Sd40f, without Sn filter): (a) the Ti prosthesis, (b) the SS prosthesis and (c) the Revitan straight hip system composed of Ti and CoCrMo. For each prosthesis, the real prosthesis (left), the 3D reconstruction (middle) and the identification material result obtained using our method (right) are shown. The yellow and blue colors correspond to the materials identified as Ti and SS, respectively.

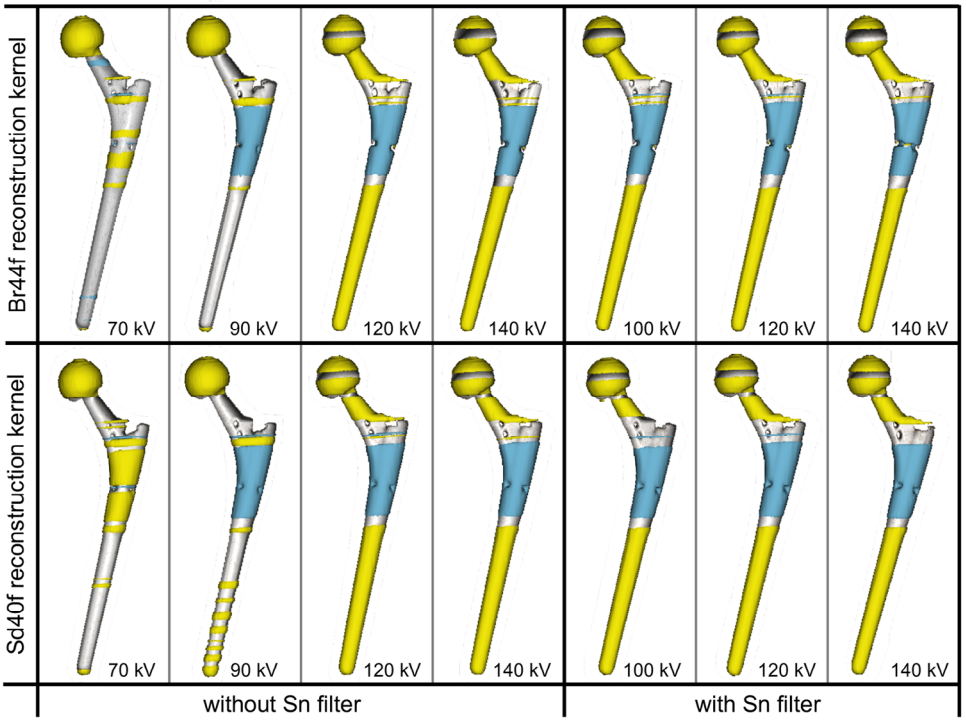


FIGURE 10 Ti and SS differentiation as a function of kV, Sn filter and reconstruction kernel for the Revitan system. The yellow and blue colors correspond to the parts of the prostheses that are identified as Ti and SS, respectively.

Increasing doses in the tissue in front of the prostheses due to electron backscatter were observed using the MC method, while this phenomenon was not observed using PBC. Similar results were obtained from Lloyd et al. when comparing the analytic anisotropic algorithm (AAA).[56] As with PBC, AAA underestimated the dose on the upstream surface of the prosthesis due to electron scatter from the high-density material. Palleri et al. compared the collapsed cone convolution (CCC) algorithm employed in Pinnacle^3 TPS (Philips, The Netherlands) to MC simulations.[57] Again, the CCC algorithm does not correctly model within 5 mm in front of the water-metal interface (−13% and −16% maximum underestimate dose were obtained for titanium and CoCr, respectively, at 18 MV). However, once the correct density was assigned to the prosthesis, the CCC algorithm succeeded in calculating the dose shadowing introduced by the prostheses with a maximum error of 3% within 0.5 cm of the high-Z cavity. Other authors have confirmed that CCC struggles with accurately calculating dose in the vicinity of a hip prosthesis.[58]

Today, it appears that the algorithms capable of accurately modeling dose near the interface of high-density objects are AXB and MC-based algorithms. Ojala et al. compared AXB with full MC and observed a peak in the proximal interface caused by the increased backscatter from the high-Z material to the lower-density material.[59] Similar to other authors,[56,60] they concluded that the

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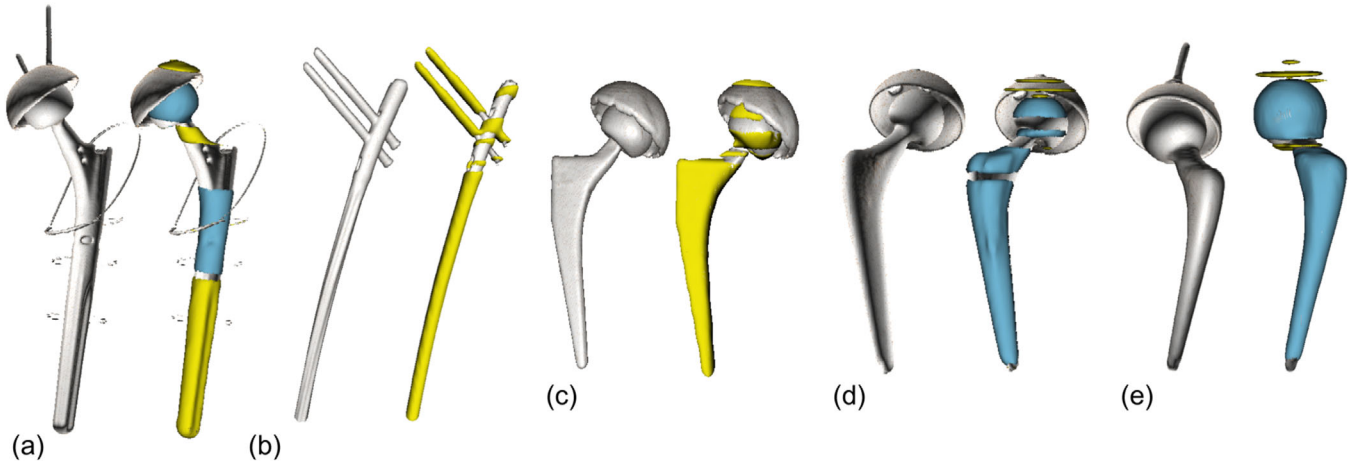


FIGURE 11 Examples of identification results obtained for five patients: (a) a Revitan system mounted with an SS head (patient 1, Sd40f kernel), (b) a femoral nail made with Ti screws (patient 5, Sd40f kernel), (c) a prosthesis with a Ti stem with a ceramic head and a Ti acetabular cup (patient 11, Br44f kernel), (d) a prosthesis with an SS stem, a CrCo head, and a Ti acetabular cup (patient 15, Sd40f kernel), and (e) a prosthesis with an SS stem and head and a Ti acetabular cup (patient 18, Br44f kernel). For each prosthesis, the prosthesis alone (left) and the material identification result obtained with our method (right) are shown. The yellow and blue colors correspond to the materials identified as Ti and SS, respectively.

TABLE 6 Summary of the detection and material differentiation results detailed in Tables 4 and 5 for energies from 110 to 140 kV.

		No. of objects			No. of parts		
		Detected	Identified correctly	Ratio [%]	Detected	Identified correctly	Comments
Sd40f	Stem/nail	1356	1352	99.7%	20/20 (100%)	20/20 (100%)	Excellent detectability. Missing slices mainly due to hip shape or partial volume effect
	Head	148	144	97.3%	18/19 (94.7%)	18/18 (100%)	Partially detected mainly due to the screw thread inside the head
	Acetab. cup	69	67	97.1%	20/20 (100%)	19/20 (95.0%)	Complex part. Only the top part, compatible with a disk shape, is visible
Br44f	**Stem/nail**	950	945	99.5%	14/14 (100%)	14/14 (100%)	Excellent detectability. Missing slices mainly due to hip shape or partial volume effect
	Head	123	113	91.9%	14/15 (93.3%)	13/14 (92.9%)	Partially detected mainly due to the screw thread inside the head
	Acetab. cup	46	43	93.5%	14/15 (93.3%)	13/14 (92.9%)	Complex part. Only the top part, compatible with a disk shape, is visible

AXB algorithm was the first commercial algorithm able to model this interface phenomenon. More importantly, with small dose deviations, on average within 2.5% when compared with MC models, beams traversing the implant are no longer prohibited due to limited dose calculation accuracy under the condition that CT images are free of artifacts.

This leads us to a second critical point, which is the accuracy of the geometry of objects with high density as well as the accuracy of HU values under these conditions. Metal artifacts in CT imaging are mainly caused by beam hardening and photon starvation and appear as shading and streaking artifacts in the reconstructed image.[17,61,62] Since the first clinical artifact reduction (MAR) algorithm based on projection interpolation,[63] various commercial MAR techniques have been proposed for metallic objects.[21,22] A very recent review[64] also showed the benefits of the MAR algorithm when coupled with virtual mono-energetic imaging using dual-energy CT (DECT). For example, the application of deep learning to MAR has recently become an active research area. In addition to MAR techniques, 16-bit reconstruction of CT data has been introduced to improve HU accuracy.[18–20] However, data indicate a high uncertainty in determining high-density metals due to cupping artifacts.[20,24] Even with advanced techniques such as eDD,[45,46] there are currently no simple procedures to correctly characterize

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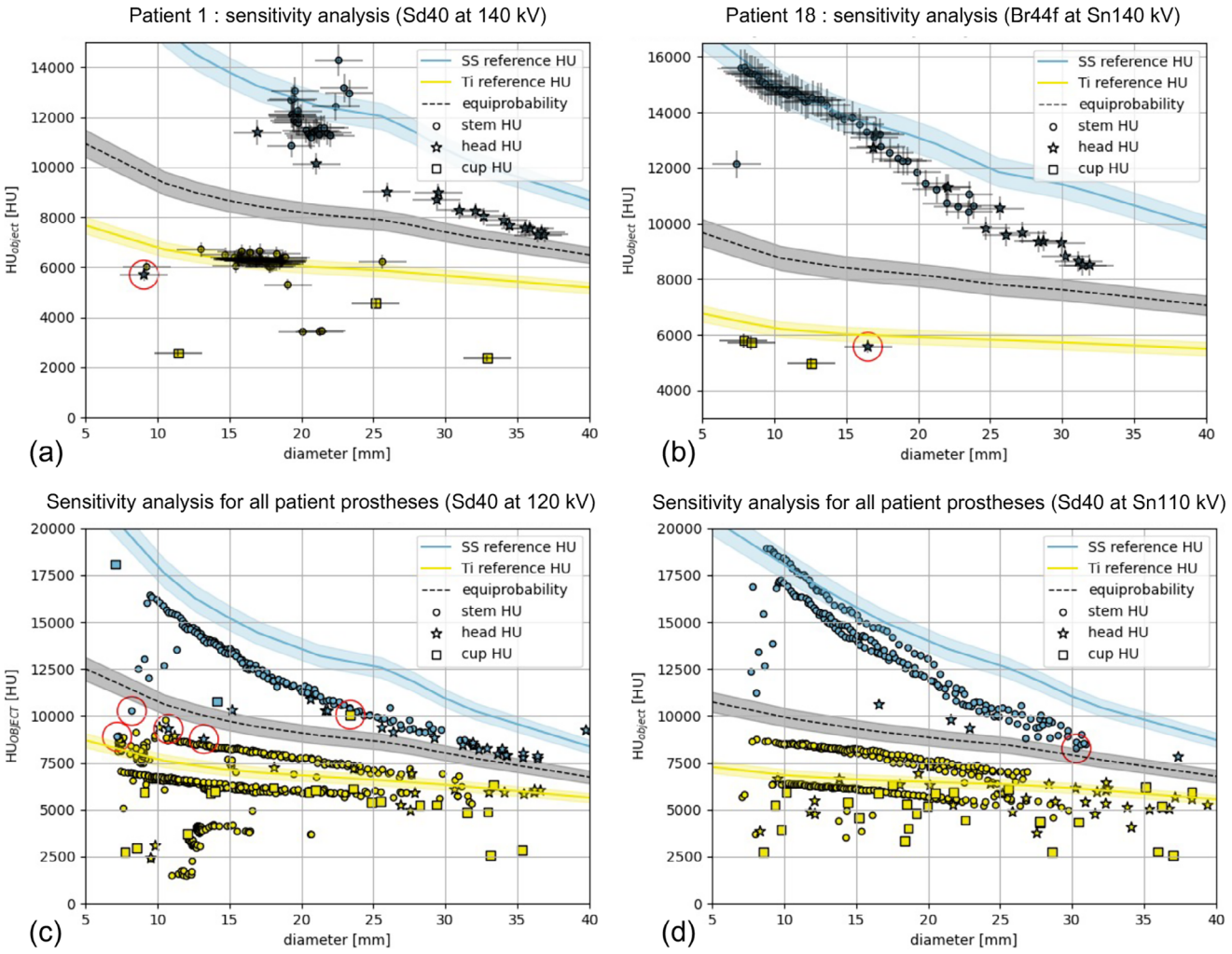


FIGURE 12 Examples of sensitivity analysis results used to validate the method: (a) a Revitan system mounted with an SS head (patient 1, Sd40f kernel), (b) a prosthesis with an SS stem and head and a Ti acetabular cup (patient 18, Br44f kernel), (c) all the patient hip prostheses acquired with the Sd40f kernel at 120 kV, and (d) all the patient hip prostheses acquired with the Sd40f kernel at Sn110 kV. The bands surrounding the curves represent the HU_{OBJECT} uncertainty ranges, while the error bars represent the uncertainties of the object diameter and the HU_{OBJECT} values. The red circles correspond to the identification errors. Note that the uncertainty bars have been omitted to make plots (c) and (d) easier to read.

metal objects on CT images. An approach, when the composition of the metallic implant remains unknown, is to use a density override value of 6 g/cm^3 (density halfway between Ti and SS) to avoid important dosimetry errors.[65] However, although simple and interesting, it does not provide a high degree of dosimetric accuracy.

In this study, we confirmed the important variation in the mean CT number with the size of high-density implants and CT spectrum (Figure 5). The ROI method used to measure the mean CT number does not appear to be robust due to the pronounced cupping artifacts. In these conditions, the extending CT-density curve for high-density metals is therefore not recommended.

The proposed method enables the material classification of different parts of hip prostheses into two groups (Ti and SS) based on measurements of reference objects. The validity range of this method depends on factors such as the reference objects used, defined validation criteria, and CT energy employed for patient scans. Although the complex shapes of the head and acetabular parts pose challenges for material differentiation, the results (Table 6) demonstrate a very high success rate in determining the material of patient hip prostheses (success $\geq$ 95% with the Sd40f reconstruction kernel) for clinical energy extending from 110 to 140 kV. Interestingly, the effect of the measurement uncertainties on the method results is low, while the shape of the prosthesis has a clear impact on the robustness of the material differentiation, as shown by the sensitivity analysis. The careful choice of the object validation criteria used in this work

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reduces the identification error rate at the expense of detectability.

This method does not require the use of complex material or dual-energy CT images. Adapting it to other CT scanner units requires only the CT data of reference objects for different energies and the reconstruction kernel used in clinical settings to compute the necessary interpolation functions described in this study.

The main limitation of our method is the need to respect the longitudinal orientation of the hip prosthesis to have a circular object shape in CT slices. This constraint may make the method more difficult to apply to other locations, such as shoulder prostheses. The second limitation is the shape of the reference objects. In this study, only cylindrical reference objects were used. To improve method accuracy, elliptical and donut shape-based reference objects could be produced to extend the reference object library. A third limitation is the number of patients enrolled in this study, which is smaller than the number of hip prosthesis solutions available on the market. However, hip prosthesis materials and shapes commercially available on the market were present in our cohort of patients. Finally, although the interpolation functions determined in our study are valid only for the Br44f and Sd40f reconstruction kernels of the go.Open Pro CT scanner unit, we are convinced that our method can be adapted and applied without difficulties to other CT scanners because the physics and types of artifacts produced by high-Z implants are the same.

5 | CONCLUSION

Extending the CT-density calibration curve can result in physical material misattribution in CT images, as there is a strong dependency on object diameter and CT spectrum for high-density metals. Therefore, automatic assignment based on extended calibration curves should not be used without major validation efforts. Because the literature shows that high-density materials used for hip prostheses can be separated into Ti and SS groups, depending on their physical density, a straight and reliable method based on interpolation functions was proposed, requiring only simple measurements on dedicated objects made of Ti and SS. Although this method relies on certain assumptions, the clinical results show excellent material detection and material differentiation for prosthesis stems. For head and acetabular cups, where the shapes are more complex, detection is partial but sufficient to allow a correct distinction between both groups. When combined with the AXB algorithm, this method improves the dosimetric accuracy of treatment plans for patients with hip prostheses.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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